

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

BIOLOGY

PAPER 3 Practical Test

4025/3

1 hour 30 minutes

SPECIMEN PAPER

Candidates answer on the question paper

Additional materials:

As listed in Instructions to Supervisors

Electronic calculator

Pencil (B or HB is recommended)

Soft clean eraser

Ruler (cm/mm)

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page.

FOR EXAMINER'S USE	
1	
2	
TOTAL	

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

This question paper consists of 8 printed pages.

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Write your answers in the spaces provided on the question paper.

Use a sharp pencil for your drawings.

*Coloured pencils and crayons should **not** be used.*

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- 1** You are required to investigate the presence of protein in four food samples.

You are provided with:

four food samples, **A**, **B**, **C** and **D**

sodium hydroxide solution

copper sulphate solution

water bath

dropper

syringe

four test tubes

benedict's solution

Proceed as follows:

1. Label **four** test tubes **A**, **B**, **C** and **D**.
2. Put 2 cm³ of sample **A** into test tube **A**.
3. Add 2 cm³ of sodium hydroxide solution into test tube **A** and shake gently.
4. Add a few drops of copper sulphate solution into test tube **A** and shake at each drop.

- (a) (i) Record any observations made in **Table 1.1**.

Table 1.1

test tube	Observations
A	
B	
C	
D	

[8]

5. Repeat the steps 2 to 4 using solutions **B**, **C** and **D** respectively.

- (ii) Suggest a precaution:

1. against biased results

2. for safety

[2]

- (iii) State **one** possible error during the procedure.

[1]

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- (iv)** Identify, with a reason, the food sample(s) that should be provided to a child suffering from kwashiorkor.

1. food sample(s)

2. reason

[3]

- (b) (i)** Design a procedure to test the food samples for reducing sugars.

[4]

- (ii)** Carry out the procedure in **(b)(i)** on food sample **A**.

Record any observations.

[2]

- 2 You are required to investigate the effect of temperature on the activity of yeast.

You are provided with the following:

2 boiling tubes
2 beakers
cold water 10 °C
hot water 40 °C
2 balloons
Spatula
thermometer
yeast mixed with glucose
string
measuring cylinder

Proceed as follows:

1. Label two beakers **A** and **B**.
2. Label two boiling tubes **A** and **B**.
3. Measure 50 cm³ of cold water and pour into beaker **A**.
4. Add a spatula of yeast and glucose mixture into beaker **A** and stir.
5. Transfer the solution to boiling tube **A**.

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6. Cover the test tube with a balloon securing with a string as shown in **Fig. 2.1**. Leave standing for five minutes.

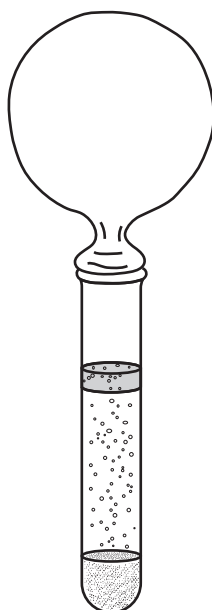


Fig. 2.1

7. Record any immediate observations and at every five minutes for fifteen minutes in **Table 2.1**.
8. Repeat steps 3 to 7 using hot water, beaker **B** and boiling tube **B**.

(a) Table 2.1

time (min)	boiling tube A	boiling tube B
0		
5		
10		
15		

[8]

- (b) (i) Identify the chemical reaction and product that caused the observations recorded in (a).

chemical reaction

product

 [2]

- (ii) State **two** reasons for stirring the contents at the start of the investigation.

 [2]

- (iii) Suggest any **two** improvements to the procedure to obtain more reliable results.

 [2]

- (c) (i) Using the results recorded in **Table 2.1**, identify the condition under which the yeast works better.

 [1]

- (ii) Identify any **two** factors that were kept constant during the experiment.

[2]

- (d) Design an experiment to test for the gas produced.

N.B. Do not carry out the experiment.

[3]

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